

EXAM 7 – SPRING 2011

13. (1.25 points)

Given the following information for a firm:

Return on equity (ROE):	10%
Book value growth rate:	5%
Discount rate:	8%

- The above values are projected to continue indefinitely.
- As of December 31, 2010, the firm's book value was \$1,000,000.

a. (0.75 point)

Use the price-to-book value ratio to estimate the value of the firm as of December 31, 2010.

b. (0.5 point)

Discuss whether it is reasonable to assume that the relationships among the values for the ROE, the book value growth rate and the discount rate are likely to continue indefinitely.

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Sample Solutions
Exam 07, 2011, Question 13

Solution 1

a)

$$ROE = 10\%$$

$$k = 8\%$$

$$BV_0 = 1,000,000$$

$$\begin{aligned} P / BV &= 1 + [(ROE - k) / (k - g)] \\ &= 1 + [(.10 - .08) / (.08 - .05)] \\ &= 1.667 \end{aligned}$$

$$P / BV \times BV = 1.667 \times 1,000,000 = 1,667,000$$

b)

It is not likely that the company will continue to see abnormal earnings indefinitely since as you continue to earn profit, this will attract more suppliers and prices will drop to remain competitive. This would lower the return until no abnormal earnings exist and returns are as expected. (Note that abnormal earnings is meant to represent earnings more than your discount rate, your expected return)